# Information is a Surprise: How do we utilize that in Ethernet?

Information is a highly perishable commodity. What computer scientists and engineers call *information* in memory and on disks might more appropriately be called *knowledge*. Bits arriving off the wire resolve the uncertainty of *when* a question is answered.

1. Once we have received (*and stored*) a bit in memory, it is no longer a surprise. This is reflected in Manchester encoding – with information conveyed by the 'transitions', not the 'levels'.
2. We claim that Information must always be *paired*: {question, binary answer}. If you send information out without having a relationship with a receiver, you are sending noise, which gets confused with entropy (the loss of information) from the transmitters' perspective. If you receive information from an unknown source that you don't have a relationship with, you are receiving noise.
3. Some notion of 'common knowledge' is required so that both ends of a channel agree on what the question/answer pairings mean *apriori*. We can think of information as the derivative of knowledge (or knowledge as the integral of information).
4. Receiving an information bit 'resolves the uncertainty' of the yes/no question. But if we remember this binary answer (by, for example, storing it in memory), then can we call it knowledge?
5. We cannot simulate quantum entanglement with a state machine with a singular serializability focus[1]. We can simulate quantum entanglement, and many of it's previously thought to be quantum-only properties (such as no-cloning and teleportation) with a Petri net and *two* conserved tokens. See Dual-SAW-Spekkens.pdf.

Maintaining liveness and synchronizing processes in network channels is a perennial challenge when packets can be dropped, reordered, duplicated or delayed. Conventional switched networks require protocol stacks and applications to use timeouts and retries to maintain liveness. This makes exactly-once semantics impossible and precipitates retry storms which lead to unbounded tail latency and transaction failure.

A superior solution is to employ individual direct channels with the Stop and Wait (SAW) protocol to maintain flow control, Colored Petri Nets1 (CPTs) to process different actions (for different orientations), and the Spekkens' Toy Model (STM) to manage epistricted consistency for distributed metadata, as described in Quantum Ethernet.

### Information Must Be Paired

One clue that information must be paired comes from the need for *disparity codes* used in Ethernet. Just as we will find 'drift' in the DC level and leakage of the gavanic isolation of networks, we will find that 'information' must also be paired when going through DC isolation elements (capacitors or transformers). Balancing rise times and fall times is insufficient, time constants may be different and more sophisticated techniques may be needed, such as in Interlaken, where the entire stream of bits are inverted. Transition Balancing is tricky enough, but relatively blunt instruments such as disparity code tricks described above are still not accurate enough to deal with precise quantities of information, such as those required in distributed transactions

But balancing ones and Zero's are also insufficient from the 'Knowledge Balance Principle' Because links break, NIC's fail, Chips fail, firmware crashes and has to be reset, and device drivers (and operating systems) fail and need to be restarted. If that was all that we had to deal with, then it would be more difficult to overcome the 'good enough' battle cry of business folk who would rather expend resources in selling what they have rather than solving a fundamental problem. This is not an unreasonable business perspective when the problem looks small (on the surface) appears to be hopeless (no solution in sight).

[1] A state machine is a Petri Net with one conserved token

But it's much worse than this: It doesn't take much thinking to recognize that the established paradigm of the networking industry, to Drop, Reorder, Duplicate and Delay packets will amplify this inability to accurately account for event pairings, such that these failures, instead of happening some $10^{-15}$, instead are happening constantly, with some large fraction of the traffic being subject to this 'information loss'. It doesn't take much imagination to see how hard it is to convert

The Interlaken Protocol: sees an XON XOFF

A key feature of Interlaken is the ability to communicate per-channel backpressure. To provide this function, two options are specified: an out-of-band flow control interface and an in-band channel. Semantically, the flow control information uses a simple on-off mechanism to signal permission to transmit on a particular channel.

There is no concept of credits with this protocol; once a channel is indicated as XON, the transmitter may send as much data as it chooses on that channel until the flow control status is changed to XOFF. The threshold whereby the receiver chooses to switch between the XON and XOFF states is a programmable option left to the user and is dependent upon the number of channels supported, depth of receive buffers, and the flow control latency of the given environment.

## Interlaken uses 64B/67B Encoding

> " An encoding/scrambling method is required for a serial interface to delineate word boundaries, provide randomness to the EMI generated by the electrical transitions, allow for clock recovery, and maintain DC balance. The encoding protocol selected for Interlaken is a modification of the 64B/66B used for the IEEE 802.3ae 10 Gigabit Ethernet specification."
>
> The existing 802.3 64B/66B solves the problem of word boundary delineation by combining a scrambled payload with two additional unscrambled bits prepended onto each 64-bit data or control word. If these sync bits are `01` they signify a data word, and if they are `10` they signify a control word; the combinations `00` and `11` are not allowed. By searching for the valid patterns in the received data stream, the receiving device declares word boundary lock after 64 correct matches, and it maintains lock by continually fixing on these two bits.
>
> Although we currently use an Intanglement Slice for the header for our protocol (because 64-bits / 8 octets is most highly probable 'set' of bits we can assume are 'atomic', and therefore a candidate for 'logically simultaneity' in our protocol.
>
> One weakness of this approach, however, is an unbounded baseline wander. Baseline wander, or DC imbalance, is caused by the accumulated excess of 1's or 0's transmitted on an individual SerDes lane. An electrical transition has an associated time constant, which in high-speed interfaces often does not allow a full voltage swing before the next bit is transmitted. Therefore, a sustained imbalance in either the number of 1's or 0's can produce a movement in the center voltage of the differential pair's eye opening. Analysis of the 64B/66B scrambler polynomial shows that over a 64Kbit time scale a running disparity in excess of +/- 1,000 bits can occur, which can produce excessive eye shifts, cause complications in the design of receiver circuitry, and increase the bit-error rate.
>
> To bound this effect, Interlaken inverts the sense of the bits in each transmitted word such that the running disparity always stays within a +/- 96-bit bound. Each lane of the bundle maintains a running count of the disparity: a '1' bit increments the disparity by one, and a '0' bit decrements the disparity by one. Before transmission, the disparity of the new word is calculated and then compared to the current running disparity. If the new word and the Table existing disparity both have the same sign, the bits within the new word are inverted. A framing bit is supplied in bit position 66 so the receiver may identify whether the bits for that word are inverted, as below:

RESEARCH: Information Surprisal:

RESEARCH. Excellent Introduction to Shannon:

## Pairing Information, Not Transitions

Unlike disparity codes and long term DC balance, information can be accounted for in pairs by setting up the system of question/answer to grow and shrink as the protocol proceeds through a transaction. We claim that information can be balanced if and only if (from Schmidt et al).

1. The system has Local Operations and Shared Randomness (LOSR) as a resource[2]. A tripartite (Alice, Bob, Charlie) setup provides only so many alternatives for orienting 'causality'.

2. Hidden Nonlocality. The system starts from a quiescent state of 'alternating causality'. The Stop and Wait (SAW) protocol provides the mechanism for both shared randomness, and serves as the 'hidden variable' in our classical (toy) quantum entanglement.

3. Network structures are critically relevant. See Figures below (Taken from Schmidt et al).

SPEKKENS Significant changes in V2: New title, new section on the relevance of network structure for entanglement quantification, new section on hidden nonlocality. More extensive discussion of the contrast between the standard definition of self-testing and our resource-theoretic LOSR-based definition, especially with regard to their divergent implications for mixed states and convexly nonextremal boxes

A standard approach to quantifying resources is to determine which operations on the resources are freely available, and to deduce the partial order over resources that is induced by the relation of convertibility under the free operations. If the resource of interest is the nonclassicality of the correlations embodied in a quantum state, i.e., entanglement, then the common assumption is that the appropriate choice of free operations is Local Operations and Classical Communication (LOCC). We here advocate for the study of a different choice of free operations, namely, Local Operations and Shared Randomness (LOSR), and demonstrate its utility in understanding the interplay between the entanglement of states and the nonlocality of the correlations in Bell experiments. Specifically, we show that the LOSR paradigm (i) provides a resolution of the anomalies of nonlocality, wherein partially entangled states exhibit more nonlocality than maximally entangled states, (ii) entails new notions of genuine multipartite entanglement and nonlocality that are free of the pathological features of the conventional notions, and (iii) makes possible a resource-theoretic account of the self-testing of entangled states which generalizes and simplifies prior results. Along the way, we derive some fundamental results concerning the necessary and sufficient conditions for convertibility between pure entangled states under LOSR and highlight some of their consequences, such as the impossibility of catalysis for bipartite pure states. The resource-theoretic perspective also clarifies why it is neither surprising nor problematic that there are mixed entangled states which do not violate any Bell inequality. Our results motivate the study of LOSR-entanglement as a new branch of entanglement theory.

## Nature has a built-in Handshake

We know from Bell state measurements that correlations can appear faster than the speed of light (one way non-local effects). But signaling of information requires a prepare-commit handshake. We take advantage of this insight in the design of an 'Information balancing' protocol for Ethernet. See: Slice-Engine.pdf and State-Machine-Specification.pdf. We observe that the well established prior art – Stop And Wait (SAW) protocol provides an 'alternating causality' where Alice is waiting for Bob's answer to her question, and Bob is waiting for Alice's answer to his question. In both cases, the question is symmetric: "are you there".

[2] Developing a new branch of entanglement theory.